Jim Griffin

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PROFILE

Computational biologist with eight years of industry experience applying and validating ML models on biological data, building reusable computational workflows for biotech R&D, and generating insights from omics datasets. Expertise in sequence-function modeling and structure prediction, along with a deep bioinformatics background. Track record spans early-stage discovery through pre-IND therapeutic development, with cross-functional experience partnering with wet-lab scientists, regulatory teams, and company leadership.

EDUCATION

Northwestern University 2013 - 2018
Ph.D. in Chemical Engineering *Evanston, IL*

Massachusetts Institute of Technology 2008 - 2012
B.S. in Chemical Engineering *Cambridge, MA*

INDUSTRY EXPERIENCE

Amber Bio 2022 - Present
Computational Biologist → Senior → Principal *South San Francisco, CA*

- Designed and optimized gene editing constructs with nucleic acid language models in PyTorch, e.g.:
    - Built and trained language models to predict sequence–function relationships from HTS data, improving screen-design efficiency and off-target prediction
    - Adapted off-the-shelf models like Pangolin/SpliceAI to improve splicing efficiency and Borzoi/AlphaGenome to improve expression and reduce methylation
    - Evaluated RNA secondary structure using structure prediction tools (ESMFold, RhoFold)
- Designed and analyzed high-throughput screens of gene editing constructs resulting in orders of magnitude increases in editing activity
- Developed NGS analysis workflows including custom Illumina RNA enrichment protocols, PacBio and Nanopore long-read RNAseq, and a variety of assays for activity, specificity and off-target profiling — and packaged them as reusable tools used across the discovery team
- Built a scalable mining platform (Docker, AWS Batch, Snakemake) that processed terabytes of public metagenomic data, discovering and patenting novel RNA-targeting CRISPR-Cas systems
- Built org-wide Claude Skills and MCP integrations connecting Claude to Benchling and S3, enabling wet-lab scientists to self-service data analysis regardless of coding ability
- Collaborated with experimental scientists to design experiments, assays and troubleshoot AAV-based gene editing vectors, leading to a poster at ASGCT 2026

Encoded Therapeutics 2020 - 2022
Bioinformatics scientist *South San Francisco, CA*

- Designed and analyzed high-throughput sequencing assays for promoter, enhancer, and transcription factor discovery screens

- Developed AAV QC assays such as a qPCR based host-cell DNA contamination assay targeting conserved high copy number retrotransposon elements in the human genome
- Performed NGS-based off-target and safety analysis (integration site analysis) for the company's preclinical candidates, and supported FDA regulatory filings (e.g. IND applications)

Ginkgo Bioworks 2018 - 2020

Computational biologist, Next Generation Sequencing (NGS) group *Boston, MA*

- Key scientific contributor on a broad range of projects, including:
  - **Lab-grown protein:** Enabled synthetic genome engineering by developing Illumina and Nanopore based strain verification pipelines to identify repetitive, high copy number, random DNA integration events in challenging hosts
  - **Live agricultural amendments:** Sourced diverse nitrogen fixation (nif) pathway sequences and performed conservation analysis to support strain and protein engineering
- Productionized Ginkgo's long-read sequencing capabilities and scaled throughput 5X via wet-lab process changes, data tracking and analysis pipeline development
- Developed Dockerized workflows in AWS for new NGS services including amplicon processing, genome assembly and annotation, comparative genomics, RNAseq, and structural variant detection
- Collaborated with experimental scientists to provide custom, on-demand bioinformatics services

ACADEMIC RESEARCH EXPERIENCE

Northwestern University 2013 – 2018

Graduate student in WeBIOME lab *Evanston, IL*

- Designed and implemented a sequence assembly, alignment and functional annotation workflow for novel microbial genomes from metagenomic samples using tools such as IDBA, BWA, samtools, bedtools, and GATK-toolkit, available at: github.com/jimbopants/Metagenomics_assembly
- Designed and built a pipeline for clustering amplicon sequencing data for use by lab members with minimal programming experience, available at: github.com/jimbopants/amplicon_processing

SELECTED PUBLICATIONS

- J Borrajo, K Javanmardi, **JS Griffin**, et al. (2023). Programmable multi-kilobase RNA editing using CRISPR-mediated trans-splicing. *bioRxiv*.
- For additional publications see: https://scholar.google.com/citations?user=z8KVusIAAAAJ

SKILLS

- Proficient in: Python, SQL, R, bash
- Scientific tooling: AWS ecosystem, Docker, Snakemake, Nextflow
- Statistics and machine learning: PyTorch, scikit-learn, and applied bio ML tools: e.g. RNA / protein folding (AlphaFold, ESMFold), foundation models and prediction tools (DNABERT, Borzoi, SpliceAI)
- Comfortable with LLM/agent-assisted programming and biological science research tools (Claude Code/Science)